# Using Databases

INTRODUCTION

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# Learning Objectives

Students must be able to:

1 define the term database

2 describe types of databases

3 explain the benefits of databases

4 describe database models

5 state features of databases

6 name examples of database management software (DBMS)

# Introduction to databases

Managing data as a resource is an important task in any organisation today. This is because data within an organisation and its environment forms part of the strategic , tactical, operational management and decision making. In order to enhance convenient and easy storage and retrieval of data, we use databases.

A database is defined as a collection of related data items organised so as to provide consistent and controlled access.

# Define the term database in your own words…

Use the diagram to help your definition.

A database is…

# Types of databases

There are two main types of databases:

Manual database

Electronic database

# Manual Database

A manual database consists if a collection of repository files and folders, each properly tagged and kept in proper sequence in a cabinet. For example, in a school the admission office stores all student file identified using unique student admission numbers such as 201/2019, 202/2019 etc. However, as organisation data and information grows manual organization becomes difficult and time consuming resulting in issues.

Can you identify some issue with type of database management?

# Manual Database

*Redundancy* – Redundancy refers to unnecessary duplication of data in multiple files

*Lack of flexibility* - It is difficult to search for a record.

*Lack of integrity* - Poor storage and maintenance of data may lead to unreliable and misleading reports.

*Lack of sharing data* – Data or information in difference manual files may not be available to other users.

# Electronic databases

Electronic database refers to the use of electronic devices such as computers to store and organise files for easy access. For example, in Microsoft Windows, files are stored under one *folder* on a storage media. Actually, the term folder as used in Windows is a metaphor that refer to the way we keep documents in a physical document wallet also referred to as a ‘document folder’.

Can you give the advantages of an electronic database?

# Electronic databases

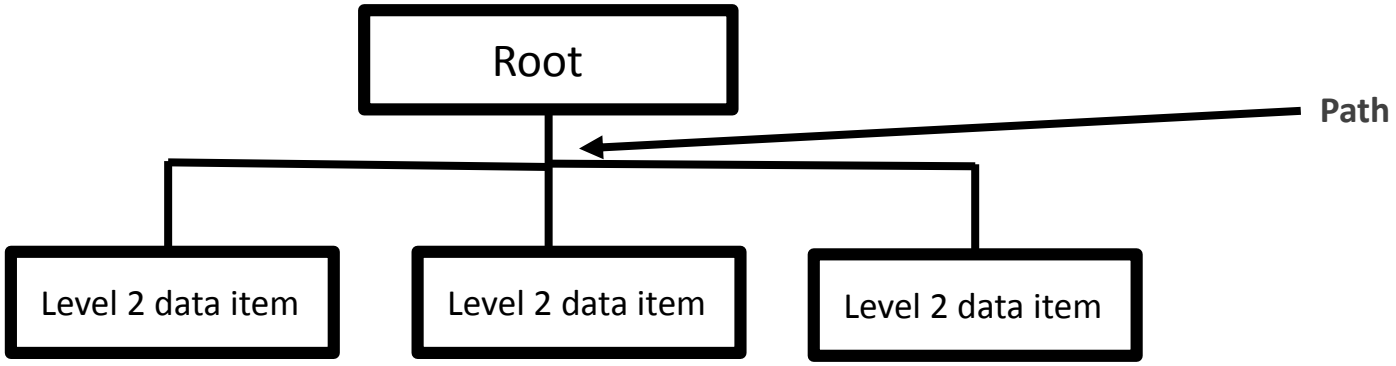
The advantage of electronic database systems provide users with tools for easy storage and access of data and information.

# Database models

A database model is a logical structure that determines how data is stored, organised and manipulated by DBMS software (Database management software). Five common types of database models are; hierarchical model, network model, relational model, object/relational model and object orientated model. The most popular model is the relational database model.

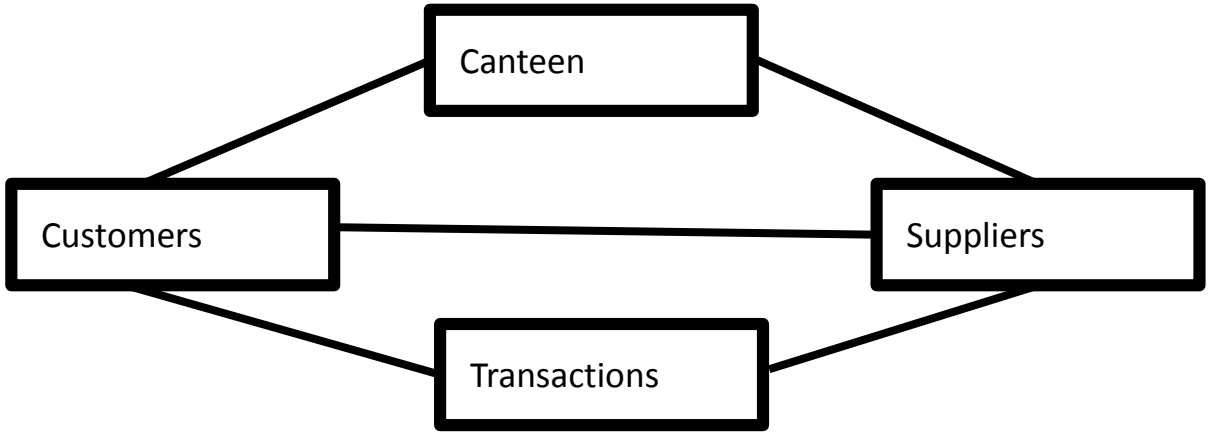
# Hierarchical Model

In the hierarchical model data is organised in hierarchical structure as seen in the image. For example to access level data, the user has to access the root first and follow the path to the level. A specific path leads to each item at lower levels. Hierarchical model is rarely used in modern database systems, hence obsolete.



# Network Model

The network model allows each record to have multiple parent and child records. This forms network structure as seen in the picture. In network models, access to a record can be through multiple paths and from any record. This type of record access structure is what is referred to as many to many relationship. Like the hierarchical model the network model is pretty much obsolete.



# Relational Model

This is the most popular type of database. This is a model where two tables can be linked together such that a record from one table references to a record or records in another table. For example Table 1 and 2 show the customer and order table that are related to each other through customer number and customer ID. In this case, the orders table shows that customer number 450 has three orders in the orders table.

| Customer number | Name | Telephone number |
|---|---|---|
| 900 | Mary Njemile | 02078907 |
| 230 | Peter Onani | 0667896 |
| 450 | Bat Masamba | 04456000 |

| Customer ID | Order ID | Date | Amount |
|---|---|---|---|
| 900 | 380 | 2/3/2014 | 90000 |
| 450 | 234 | 2/3/2014 | 5000 |
| 450 | 123 | 2/4/2014 | 12000 |
| 450 | 667 | 4/4/2014 | 15000 |

# Object relational Model

In software development, there is a shift from structured programming to object-orientated programming an object is a representation of real world using state and behaviour. This shift raised need for database model that stores and manipulates objects. Therefore object-relational is an evolutionary hybrid model that has capabilities to handle both relational and object orientated storage and manipulation of data

# Object orientated Model

Object orientated models purely support storage, access and manipulation of objects. Although currently not popular in the market, object orientated database management systems provides unification of application and database development. This is different from current approaches in RDBMS where a separate application has to be developed to access a database.

# Database structure

For easy storage, retrieval and manipulation, RDBMS organise data into a structure called a schema. A schema is an organisation structure made up of fields at the lowest level and database at the highest level.

# Database structure definitions

Field – A field is a set of characters or logical combination of characters that defines a single data value

Record – A record is a collection of related fields that represent a single data item.

Table – A table is the equivalent of a file and contains a set of related records.

Database – A database is a collection of related tables and each table consists of related records made up of fields. This type of organisation is what is referred to as a database schema

**Question –** State the five types of database models.